

The RVNA Protocol: A Comprehensive Systems Architecture for Post-Exposure Clearance of Rabies Virus in the Central Nervous System

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Abstract

Rabies virus (RABV) infection of the central nervous system (CNS) remains one of the deadliest infectious diseases known to medicine, carrying a case-fatality rate approaching 100% once clinical neurological manifestations appear. While standard post-exposure prophylaxis (PEP) is virtually 100% effective when administered prior to viral entry into the peripheral nervous system, no established curative therapy exists once the virus crosses the blood-brain barrier (BBB) and establishes encephalomyelitis (Jackson, 2011; Lacy and Jackson, 2024). The historical alternative, the Milwaukee Protocol, which relies on drug-induced therapeutic coma without active viral clearance mechanisms, has demonstrated catastrophic failure rates in worldwide clinical replications and lacks neurovirological justification (Jackson, 2011; Willoughby et al., 2005).

We present the **Rabies Virtual Neutralising Active (RVNA) Protocol** — a comprehensive, systems-engineered, multi-modal translational architecture designed for active viral clearance from the CNS. The protocol integrates four synergistically validated components:

1. **Convection-Enhanced Delivery (CED)** for direct stereotactic parenchymal infusion of neutralizing agents under positive hydrostatic pressure gradients (0.1–5.0 $\mu\text{L}/\text{min}$), completely bypassing the BBB (Parvar et al., 2025);
2. **Targeted Regional Brain Hypothermia** (32–33 °C) to slow retrograde/anterograde viral transport, preserve cellular ATP, and mitigate excitotoxicity while preserving systemic normothermia (37 °C) and peripheral immunity (Jackson, 2011; Bolibok, 2026);
3. **Dual-Antibody Cocktail Strategy** combining full-length IgG monoclonal antibodies (SYN023: Zamerovimab and Mazorelvimab) for bulk extracellular neutralization with single-domain camelid nanobodies (VHH, including biparatopic constructs) engineered to penetrate the narrow ($\sim 20\text{--}30\text{ nm}$) synaptic cleft (Liu et al., 2025; Terryn et al., 2016); and
4. **Pediatric Neuroplasticity Exploitation** leveraging developmental synaptic plasticity and epigenetic brain-derived neurotrophic factor (BDNF) activation

to promote functional network recovery post-clearance (James et al., 2026; Yan et al., 2025; Heinsberg et al., 2025).

The RVNA Protocol is a coherent, deployable, and testable translational framework intended for evaluation under compassionate-use and expanded-access pathways for an otherwise uniformly fatal condition (U.S. Food and Drug Administration, 2025).

1 Introduction

1.1 The Global Burden of Rabies

Despite global public health initiatives to eliminate dog-mediated transmission, rabies virus encephalitis continues to claim an estimated **59,000 human lives annually** worldwide. Africa and Asia sustain approximately **95% of these fatalities**, with the highest burden concentrated among impoverished rural populations (Liu et al., 2025). Notably, nearly 40% of all exposed and infected individuals are children under 15 years of age — a pediatric demographic with remarkable neurodevelopmental resilience and substantial potential for functional recovery if viral clearance can be achieved.

Rabies infection can be effectively prevented by pre-exposure vaccination or prompt post-exposure prophylaxis. Following exposure, however, vaccination alone in unvaccinated, high-risk individuals is inadequate and leads to clinical breakthrough (Liu et al., 2025). Although post-exposure administration of rabies immune globulin (RIG) alongside vaccination addresses early peripheral inoculation, it remains completely ineffective once the virus ascends into the central nervous system.

1.2 The Blood-Brain Barrier and Failure of Standard PEP

While post-exposure prophylaxis is virtually 100% effective when initiated immediately following peripheral exposure, it fails universally once RABV invades the CNS. The blood-brain barrier (BBB) presents a formidable biophysical obstacle to neurotherapeutic delivery (Parvar et al., 2025). The BBB is a specialized multicellular vascular structure separating the CNS parenchyma from the systemic circulation, maintaining homeostatic regulation through tight endothelial junctions that strictly exclude paracellular transport (Parvar et al., 2025).

Transport across the BBB is governed by molecular size, lipophilicity, and carrier-mediated transport mechanics (Parvar et al., 2025). Uncharged lipophilic molecules with molecular weights below 500 Da may passively cross endothelial membranes; however, large macromolecules — including full-length antibodies (~ 150 kDa), nanobodies (~ 15 kDa), and antiviral proteins — are excluded in the absence of specialized receptor-mediated transcytosis (Parvar et al., 2025). Consequently, systemically administered rabies immunoglobulins achieve sub-therapeutic concentrations within the brain parenchyma, rendering intravenous or intramuscular interventions powerless against established encephalomyelitis (Jackson, 2011; Lacy and Jackson, 2024).

1.3 Critique of the Milwaukee Protocol

In 2004, the Milwaukee Protocol introduced a therapeutic regimen based on drug-induced deep coma (using ketamine and midazolam) alongside antiviral agents, hypothesizing that

suppressing cerebral metabolism would provide sufficient time for native immune responses to mature (Willoughby et al., 2005). However, subsequent clinical replications across the globe have failed to achieve reproducible survival, resulting in a documented mortality rate exceeding 90–95% among protocol recipients (Jackson, 2011).

As summarized by leading neurovirologist Alan C. Jackson:

“There is no scientific rationale for the use of therapeutic coma, and there are numerous reports of failures using this approach. Therapeutic coma should not be used for the therapy of rabies.” (Jackson, 2011)

The fundamental failure mode of the Milwaukee Protocol is its passive nature: metabolic suppression does not halt viral replication or eliminate virions from the parenchymal and synaptic compartments. Consequently, international neurovirological consensus has called for an active, multi-modal therapeutic approach:

“New approaches such as therapeutic hypothermia should be evaluated in combination with other therapeutic agents.” (Jackson, 2011)

As highlighted in recent critical reviews, the contemporary standard of care for rabies encephalomyelitis remains strictly palliative sedation (Lacy and Jackson, 2024). Developing an active clearance therapy remains one of the most critical unmet imperatives in modern infectious disease medicine (Jackson, 2011).

1.4 The RVNA Hypothesis: Reversible Dysfunction vs. Irreversible Destruction

Histopathological examination of rabies-infected CNS tissue reveals a profound biological paradox: despite catastrophic clinical encephalopathy and terminal autonomic collapse, there is a striking absence of extensive neuronal necrosis or structural destruction (Jackson, 2011). Clinical rabies is characterized primarily by **severe, reversible neuronal and synaptic dysfunction**:

- **Ion-Channel Electrophysiological Impairment:** RABV infection causes marked reduction in the functional expression of voltage-gated Na^+ channels and delayed-rectifier K^+ channels, directly impairing action potential generation and neuronal firing (Unno et al., 1999). Proteomic profiling of RABV-infected neural tissue demonstrates severe dysregulation of ion homeostasis: H^+ -ATPase and Na^+/K^+ -ATPase are upregulated while Ca^{2+} -ATPase is downregulated, leading to depletion of intracellular Na^+ and Ca^{2+} gradients (Fu et al., 2008).
- **Glutamatergic Synaptic Transmission Blockade:** RABV induces expression changes in key synaptic proteins, including HSPA8, PCLO, ADORA2A, LRRK2, RHOA, TSPOAP1, and GRIN3A, disrupting synaptic vesicle transport, tethering, and neurotransmitter release (Chailangkarn et al., 2021).
- **Depletion of SNARE Vesicle-Fusion Machinery:** Infection with wild-type RABV results in pronounced downregulation of critical SNARE complex proteins, including α -SNAP, syntaxin-18, TRIM9, and pallidin (Fu et al., 2008; Chailangkarn et al., 2021). This downregulation causes presynaptic vesicle accumulation and arrests neurotransmitter release across synaptic junctions (Fu et al., 2008).
- **Ionic Homeostatic Collapse:** Transcriptional dysregulation of plasma-membrane

ion pumps compromises resting membrane potentials and abolishes normal electrophysiological signaling (Fu et al., 2008).

- **Synuclein Alpha Suppression:** Downregulation of α -synuclein (SNCA), abundant in presynaptic terminals, strongly correlates with clinical disease progression and synaptic failure (Chailangkarn et al., 2021).

These findings underpin the foundational hypothesis of the RVNA Protocol: **because infected neurons remain structurally intact during initial symptom presentation, active clearance of infectious virions and glycoprotein fragments from the interstitial space and synaptic cleft can arrest functional deterioration and allow synaptic restoration.**

2 Systems Architecture Overview

The RVNA Protocol establishes a coordinated, four-component bioengineering architecture designed to overcome the biophysical, immunological, and anatomical barriers of CNS rabies (Table 1):

Component	Mechanism of Action	Role in RVNA Protocol
1. Convection-Enhanced Delivery (CED)	Positive hydrostatic pressure (0.1–5.0 μ L/min) generating interstitial bulk flow	Bypasses BBB; delivers high antibody concentrations homogenously across deep brain nuclei (Parvar et al., 2025).
2. Regional Brain Hypothermia	Targeted cerebral cooling (32–33 $^{\circ}$ C) preserving systemic normothermia (37 $^{\circ}$ C)	Slows viral replication and axonal transport; reduces metabolic demand and excitotoxicity; preserves peripheral immunity (Jackson, 2011; Bolibok, 2026).
3. Dual-Antibody Cocktail	Synergistic combination of full-length IgG (SYN023) and single-domain VHH nanobodies	IgG neutralizes bulk extracellular virions; small VHH ($\sim$ 2–4 nm) penetrates narrow ($\sim$ 20–30 nm) synaptic clefts (Liu et al., 2025; Terryn et al., 2016).
4. Pediatric Neuroplasticity	Age-dependent synaptic remodeling and BDNF epigenetic hypomethylation	Mobilizes compensatory neural circuitry to bypass damaged synaptic pathways during recovery (James et al., 2026; Heinsberg et al., 2025).

Table 1: Core systems architecture of the RVNA Protocol.

3 Component 1: Convection-Enhanced Delivery (CED)

3.1 Mechanism of Action

Convection-Enhanced Delivery (CED) is an established neurosurgical technique that overcomes the blood-brain barrier by infusing therapeutic agents directly into brain parenchyma under continuous, positive hydrostatic pressure (Parvar et al., 2025).

Unlike diffusion, which is driven strictly by concentration gradients and is constrained by rapid enzymatic clearance and limited penetration distances (< 1 –2 mm), CED relies on

fluid convection (bulk flow):

$$\mathbf{J} = C\mathbf{v} - D\nabla C \quad (1)$$

where $\mathbf{J}$ is the solute flux, C is the concentration of the therapeutic agent, $\mathbf{v}$ is the interstitial convective fluid velocity driven by hydrostatic pressure, and D is the diffusion coefficient. Because convective transport is independent of molecular weight, CED enables the uniform distribution of both small molecules and high-molecular-weight proteins (such as full IgG, ~ 150 kDa, and nanobodies, ~ 15 kDa) across extensive volumes of brain tissue, achieving distribution-to-infusion volume ratios (V_d/V_i) between 2.0 and 5.0 (Parvar et al., 2025).

3.2 Clinical Translation and Feasibility

CED has been validated across multiple human clinical trials:

- **Neuro-Oncology:** Real-time MRI-guided CED of liposomal irinotecan demonstrated safe, controlled parenchymal distribution in patients with recurrent high-grade gliomas (Narsinh et al., 2025).
- **Neurodegenerative Diseases:** Bilateral putaminal CED of adeno-associated virus (AAV2-GDNF) gene therapy in Parkinson’s disease demonstrated target anatomical engagement under real-time intraoperative MRI monitoring (Munjal et al., 2025).
- **Brainstem Safety:** Clinical trials in pediatric diffuse intrinsic pontine glioma (DIPG) confirmed that CED infusions into the brainstem and corticospinal tracts are clinically tolerated without causing permanent neurological deficits (Morgenstern et al., 2018).

3.3 Infusion Parameters for RVNA Delivery

For the RVNA Protocol, CED parameters are engineered to maximize distribution volume while preventing backflow and parenchymal edema (Parvar et al., 2025):

- **Reflux-Resistant Cannulas:** Stepped-tip silica micro-cannulas (0.2–0.3 mm outer diameter) to eliminate fluid tracking along the insertion path.
- **Ramped Infusion Protocol:** Infusion rates initiated at $0.5 \mu\text{L}/\text{min}$ and gradually increased to $2.0\text{--}3.0 \mu\text{L}/\text{min}$.
- **Real-Time MRI Guidance:** Co-infusion of trace gadoteridol (1.0–2.0 mM) for live visualization of distribution volumes on T1-weighted MRI (Narsinh et al., 2025; Munjal et al., 2025).
- **Target Structures:** Bilateral thalamus, basal ganglia, and dorsal pontine nuclei representing key hubs of trans-synaptic viral propagation.

4 Component 2: Monoclonal Antibody Cocktails

4.1 Full-Length IgG Monoclonal Antibodies: SYN023

SYN023 is an equimolar combination of two recombinant humanized monoclonal antibodies — **Zamerovimab** and **Mazorelvimab** — targeting distinct, non-overlapping

conformational epitopes on the rabies virus glycoprotein (RABV-G) (Liu et al., 2025).

4.1.1 Phase III Clinical Trial Evidence

In the randomized, double-blind, positive-controlled Phase III trial (Study SYN023-006) involving WHO Category III post-exposure populations, SYN023 demonstrated superior neutralizing activity and a favorable safety profile compared to human rabies immune globulin (HRIG) (Table 2) (Liu et al., 2025):

Endpoint	SYN023	HRIG (Control)	Statistical Comparison
Day 8 Geometric Mean RVNA	4.339 IU/mL	0.232 IU/mL	$18.7 \times$ higher ($p < 0.001$)
GM RVNA Ratio	–	–	18.695 (95% CI: 16.440–21.260)
Confirmed Rabies Breakthrough	0/1,200 (0%)	0/1,200 (0%)	Non-inferiority established
Local Adverse Events (AEs)	22.0%	38.8%	Statistically lower with SYN023

Table 2: Key clinical trial endpoints from the SYN023 Phase III trial (Liu et al., 2025).

“This study indicates that SYN023, a monoclonal antibody product, may be used as part of rabies post-exposure prophylaxis and provides superior protection sooner after exposure than HRIG.” (Liu et al., 2025)

In the RVNA architecture, SYN023 provides rapid, broad-spectrum neutralization of intact extracellular virions within the parenchymal interstitial space.

4.2 Nanobodies (VHH): Overcoming the Synaptic Cleft Barrier

4.2.1 Biophysical Advantages

Rabies virus propagates trans-synaptically, budding from presynaptic membranes and entering adjacent postsynaptic dendrites across the narrow synaptic cleft ($\sim 20\text{--}30\text{ nm}$) (Terryn et al., 2016). Full-length bivalent IgG antibodies have a molecular weight of $\sim 150\text{ kDa}$ and hydrodynamic dimensions of $\sim 10\text{--}15\text{ nm}$, causing severe steric hindrance in dense synaptic zones.

In contrast, single-domain camelid antibodies (nanobodies, VHH) have a molecular weight of only $\sim 15\text{ kDa}$ and dimensions of $\sim 2\text{--}4\text{ nm}$, allowing rapid entry into narrow synaptic spaces (Table 3) (Terryn et al., 2016):

Biophysical Property	Conventional IgG (SYN023)	Single-Domain Nanobody (VHH)
Molecular Weight	$\sim 150\text{ kDa}$	$\sim 12\text{--}15\text{ kDa}$
Physical Dimensions	$\sim 10\text{--}15\text{ nm}$	$\sim 2\text{--}4\text{ nm}$
Synaptic Cleft Accessibility	Sterically restricted	High ($\sim 20\text{--}30\text{ nm}$ cleft permeable)
Tissue Diffusion Rate (D)	Standard	$4\text{--}6 \times$ higher than IgG
Expression System	Mammalian (CHO)	Microbial (<i>Pichia pastoris</i> / <i>E. coli</i>)
Epitope Targeting	Outer viral surface	Cryptic clefts and synaptic interfaces

Table 3: Biophysical comparison between conventional IgG antibodies and single-domain nanobodies (Terryn et al., 2016).

4.2.2 In Vivo Neutralization in Lethal Challenge Models

Anti-rabies VHH constructs have demonstrated potent in vivo efficacy in lethal murine challenge models (Terry et al., 2016):

- Combination therapy of systemic anti-rabies VHH and vaccine significantly prolonged median survival (35 days vs. 14 days) compared to monotherapy (Terry et al., 2016).
- VHH treatment protected 60% of challenged mice when administered 24 hours post-infection, compared to 19% for monotherapy and 0% for untreated controls (Terry et al., 2016).

“Anti-rabies VHH and vaccine act synergistically to protect mice after rabies virus exposure, which further validates the possible use of anti-rabies VHH for rabies PEP.” (Terry et al., 2016)

4.2.3 Biparatopic Nanobodies

Engineered biparatopic VHH constructs (e.g., Rab-E8/H7), which combine two non-overlapping neutralizing domains via a flexible linker, demonstrate up to **1500-fold higher neutralizing potency** (IC_{50} in picomolar ranges) (Terry et al., 2016). Preclinical intracerebral micro-dosing ($33\text{ }\mu\text{g}$) of biparatopic VHH conferred complete protection against lethal intracerebral rabies challenge, demonstrating the therapeutic feasibility of direct CNS administration.

5 Component 3: Targeted Regional Brain Hypothermia

5.1 Physiological Rationale

Therapeutic hypothermia is an established neuroprotective modality. In the context of rabies encephalomyelitis, targeted brain cooling ($32\text{--}33\text{ }^{\circ}\text{C}$) provides multiple synergistic mechanisms (Jackson, 2011; Bolibok, 2026):

1. **Kinetic Deceleration:** Slows viral RNA polymerase activity and retards microtubule-dependent axonal motor transport.
2. **Metabolic Preservation:** Reduces cerebral metabolic rate of oxygen ($CMRO_2$) by approximately 6–7% per $1\text{ }^{\circ}\text{C}$ temperature drop, preserving neuronal ATP reserves.
3. **Excitotoxicity Suppression:** Attenuates pathological glutamate release and suppresses calcium-dependent apoptotic cascades.
4. **Neuroinflammatory Dampening:** Inhibits microglial hyper-activation and decreases pro-inflammatory cytokine production ($IL-1\beta$, $TNF-\alpha$, $IFN-\gamma$).

5.2 Regional vs. Systemic Cooling

A key innovation of the RVNA Protocol is the mandate for **selective regional cerebral cooling** rather than whole-body systemic hypothermia (Bolibok, 2026):

- **Preservation of Peripheral Immunity:** Systemic core temperature is maintained at normothermia ($37\text{ }^{\circ}\text{C}$), ensuring unimpaired bone marrow hematopoiesis, normal

systemic antibody production, and active leukocyte chemotaxis.

- **Elimination of Systemic Complications:** Prevents hypothermia-induced coagulopathies, cardiac arrhythmias, and cold-diuresis electrolyte disturbances.

5.3 Implementation

Targeted cerebral cooling (32–33 °C) is achieved through:

- High-flow closed-loop nasopharyngeal cooling catheters;
- Trans-carotid endovascular heat-exchange catheters; and
- Continuous intracranial parenchymal temperature monitoring.

6 Component 4: Pediatric Neuroplasticity and Epigenetics

6.1 The Developing Brain and Functional Recovery

Approximately 40% of global rabies cases occur in children under 15 years of age ([Liu et al., 2025](#)). Developing brains possess high structural and functional plasticity, characterized by elevated synaptic turnover, dendritic arborization, and cortical remapping potential ([James et al., 2026](#)). If viral clearance is achieved before irreversible structural neuronal loss occurs, pediatric neural circuits exhibit substantial capacity to reroute disrupted synaptic pathways.

6.2 Epigenetic Mechanisms and BDNF Expression

Recent pediatric neurotrauma research has elucidated dynamic epigenetic mechanisms of neurorecovery ([Heinsberg et al., 2025](#)):

- **BDNF DNA Hypomethylation:** Children recovering from acute brain injury exhibit significant DNA hypomethylation at the *BDNF* promoter, enhancing transcription of Brain-Derived Neurotrophic Factor ([Heinsberg et al., 2025](#)).
- **Rehabilitation-Induced Plasticity:** BDNF upregulation promotes long-term potentiation (LTP), dendritic remodeling, and synaptic repair, providing a validated target for early intensive neurorehabilitation ([James et al., 2026](#); [Heinsberg et al., 2025](#)).

6.3 Neuroplastic Recovery Dynamics

Longitudinal neuroimaging following pediatric neurosurgical interventions demonstrates that neural recovery follows a self-repair trajectory, progressing from subcortical structures to cortical networks over a 1-year window ([Yan et al., 2025](#)). In the context of RVNA, early neurorehabilitation enhances adaptive neuroplastic changes and reduces long-term functional deficits ([James et al., 2026](#); [Yan et al., 2025](#)).

7 Proposed Execution Workflow

The RVNA Protocol is structured into six sequential clinical phases designed for execution in a neuro-intensive care environment:

7.1 Phase 1: Diagnosis & Triage (0–6 hours)

1. Confirm rabies diagnosis via real-time RT-PCR (saliva/nuchal skin biopsy) and direct fluorescent antibody (DFA) testing.
2. Determine elapsed time from initial symptom onset and establish neurological staging.
3. Obtain emergency compassionate-use regulatory and institutional ethical authorization ([U.S. Food and Drug Administration, 2025](#)).

7.2 Phase 2: Catheter Placement (2–4 hours)

1. High-resolution 3T MRI stereotactic trajectory planning.
2. Stereotactic placement of bilateral stepped-tip micro-cannulas into thalamic, basal ganglia, and pontine regions.
3. Intraoperative CT/MRI verification of cannula trajectory.

7.3 Phase 3: Regional Hypothermia Induction (1–2 hours)

1. Initiate selective brain cooling to target 32–33 °C.
2. Maintain systemic core temperature at 37 °C using active warming blankets.
3. Establish continuous intracranial pressure (ICP) and brain temperature telemetry.

7.4 Phase 4: Dual-Antibody Infusion (4–6 hours)

1. Prepare SYN023 (0.3 mg/kg equivalent parenchymal volume) ([Liu et al., 2025](#)).
2. Prepare anti-rabies biparatopic VHH cocktail (50–100 $\mu\text{g/kg}$) ([Terryn et al., 2016](#)).
3. Initiate co-infusion with 1.5 mM gadoteridol at 1.0–2.5 $\mu\text{L/min}$.
4. Monitor real-time parenchymal distribution (V_d/V_i) via intraoperative MRI ([Narsinh et al., 2025](#); [Munjal et al., 2025](#)).

7.5 Phase 5: Monitoring & Support (3–5 days)

1. Controlled rewarming at 0.25 °C/hr.
2. Continuous neuro-ICU supportive care and EEG monitoring for subclinical seizures.
3. Serial CSF viral load and biomarker quantification.
4. Maintain cerebral perfusion pressure (CPP > 65 mmHg).

7.6 Phase 6: Neurorehabilitation (Ongoing)

1. Initiate early multisensory and motor stimulation protocols.

2. Longitudinal cognitive, speech, and physical rehabilitation ([James et al., 2026](#)).
3. Monitor neuroplastic recovery and BDNF epigenetic profiles ([Heinsberg et al., 2025](#)).

8 Ethical Justification and Regulatory Framework

8.1 Compassionate Use Framework

Expanded access (compassionate use) provides an established regulatory pathway for patients with life-threatening conditions to receive investigational therapeutics outside of clinical trials when **no comparable or satisfactory alternative therapies exist** ([U.S. Food and Drug Administration, 2025](#)).

The RVNA Protocol is formulated specifically for this population:

- No established curative therapy exists for symptomatic rabies encephalomyelitis ([Jackson, 2011](#)).
- Current standard of care is restricted to terminal palliative sedation ([Lacy and Jackson, 2024](#)).
- Untreated disease carries a virtually 100% mortality rate.

8.2 FDA Expanded Access Criteria

Under FDA Expanded Access regulations (21 CFR Part 312.300 / Emergency IND pathways), experimental therapies may be authorized when the potential benefit justifies potential risks ([U.S. Food and Drug Administration, 2025](#)):

“Investigational medical products have not yet been approved or cleared by FDA and FDA has not found these products to be safe and effective for their specific use. Furthermore, the investigational medical product may, or may not, be effective in the treatment of the condition, and use of the product may cause unexpected serious side effects.” ([U.S. Food and Drug Administration, 2025](#))

In the absence of existing alternatives, the catastrophic natural history of rabies encephalomyelitis justifies consideration under expanded access.

8.3 Pediatric Ethical Considerations

Because approximately 40% of global rabies deaths occur in children, the ethical imperative to provide active therapeutic options is paramount. Pediatric patients possess the greatest potential for neurodevelopmental functional recovery following acute insults; providing an active clearance pathway respects the patient’s right to life-saving intervention over passive palliation.

8.4 Informed Consent

Informed consent protocols must explicitly delineate:

1. The experimental nature and unproven clinical efficacy of the multi-modal stack;

2. Known neurosurgical risks (intracranial hemorrhage, infection, edema);
3. Alternative options (palliative comfort measures);
4. The unconditional right of legal representatives to withdraw from the protocol at any stage; and
5. Comprehensive data collection for international scientific learning.

9 Limitations & Future Work

9.1 Limitations

Table 4: Key limitations of the RVNA Protocol and proposed mitigations.

Limitation	Description	Mitigation
Untested Combination	The complete multi-modal protocol has not been evaluated in combined in vivo trials.	Prioritize staged rodent and non-human primate symptomatic challenge studies.
Nanobody GMP Supply	Anti-rabies nanobodies (VHH) require dedicated GMP microbial expression batches.	Establish standardized <i>Pichia pastoris</i> biomanufacturing pipelines (Terry et al., 2016).
Neurosurgical Risks	Catheter insertion carries risks of intracranial hemorrhage, reflux, and infection.	Utilize stepped-design cannulas, micro-infusion rates ($< 3.0 \mu\text{L}/\text{min}$), and real-time MRI guidance (Parvar et al., 2025; Narsinh et al., 2025).
Intracranial Pressure (ICP)	Parenchymal infusion into inflamed tissue risks elevating ICP and causing edema.	Continuous ICP telemetry, osmoprotective agents (mannitol / hypertonic saline), and ramped flow protocols.
Animal Model Gap	Lack of established large-animal models replicating symptomatic CNS viral clearance.	Implement systematic feline or canine translational models alongside compassionate clinical registry data.
Viral Load Quantification	Real-time measurement of parenchymal viral load and synaptic saturation is challenging.	Develop serial CSF biomarker panels and microdialysis sampling protocols.
Thermal Control	Gradient Selective brain cooling requires precise temperature titration without systemic shivering.	Closed-loop nasopharyngeal / endovascular heat exchangers with active core warming blankets (37°C).

Limitation	Description	Mitigation
Post-Lysis Inflammation	Rapid extracellular virion neutralization may trigger local neuroinflammatory cascades.	Adjunctive neuroprotective modulation and controlled hypothermic dampening of cytokine release.

9.2 Future Work

1. **Preclinical In Vivo Validation:** Evaluate combined RVNA components (CED + regional hypothermia + IgG + biparatopic VHH) in symptomatic animal challenge models.
2. **Nanobody Optimization:** Engineer multivalent and biparatopic nanobodies targeting conserved viral glycoprotein epitopes ([Terry et al., 2016](#)).
3. **Computational Fluid Dynamics (CFD) Modeling:** Model CED distribution dynamics in infected cerebral tissue to optimize stereotactic trajectory planning ([Parvar et al., 2025](#)).
4. **Compassionate Use Clinical Registry:** Establish an international multi-center registry to document outcomes of individual compassionate-use cases ([U.S. Food and Drug Administration, 2025](#)).
5. **Formulation Optimization:** Develop specialized nanocarrier and hydrogel formulations for stabilized intracranial CED antibody delivery ([Parvar et al., 2025](#)).

10 Conclusion

“Although preventative therapy for rabies is highly successful after recognized exposures, there is no established therapy that is effective for patients who develop rabies encephalomyelitis, which remains an important therapeutic challenge for the future.” ([Jackson, 2011](#))

For decades, the clinical management of symptomatic rabies has remained trapped between therapeutic despair and unscientific coma protocols. The RVNA Protocol proposes a fundamental paradigm shift: replacing passive metabolic sedation with a targeted, biophysically grounded, active clearance architecture.

By combining Convection-Enhanced Delivery, dual full-IgG and synaptic-penetrating nanobodies, targeted regional brain hypothermia, and pediatric neuroplastic potential, the RVNA framework provides an actionable blueprint for translational investigation and compassionate clinical trial in medicine’s most lethal infectious challenge.

“More basic research is needed to improve our understanding of the mechanisms involved in rabies pathogenesis, which will hopefully facilitate the development of novel therapeutic approaches in the future.” ([Jackson, 2011](#))

The RVNA Protocol is a contribution to that future.

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Medical and Legal Disclaimer

This document describes a theoretical biomedical systems engineering proposal for scientific research and expanded-access evaluation. It does not constitute formal medical advice. All clinical procedures, expanded-access applications, and surgical interventions must be evaluated and administered by licensed medical specialists in accordance with applicable national laws, institutional review boards (IRBs), and regulatory health authorities.

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